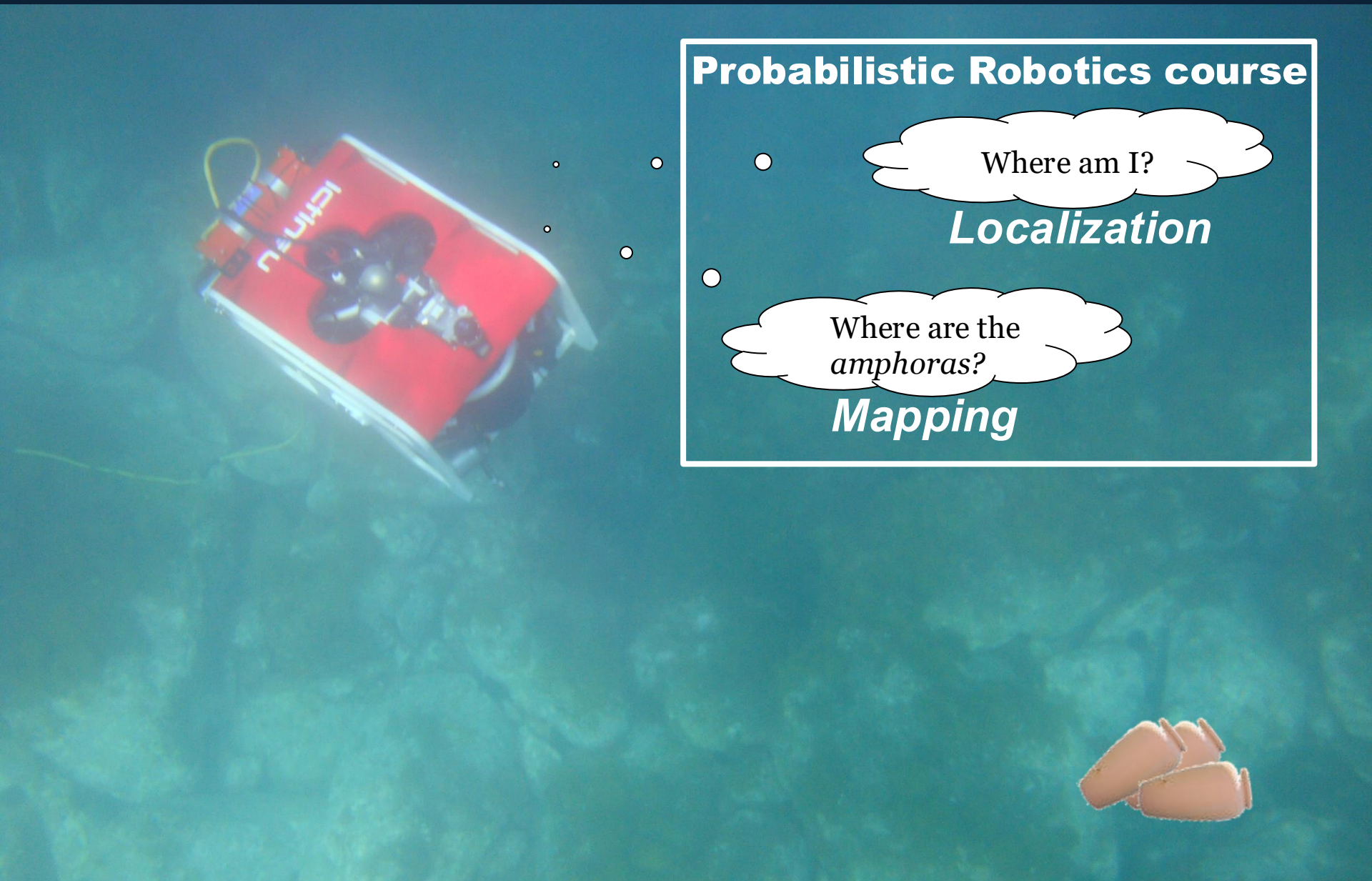
What is Probabilistic Robotics?

*'**Robotics** is the science of perceiving and manipulating the physical world through computer controlled mechanical devices.'*

*'**Probabilistic robotics** is a relatively new approach to robotics that pays tribute to the uncertainty in robot perception and action.'*

Sebastian Thrun

Probabilistic Robotics



Probabilistic Robotics



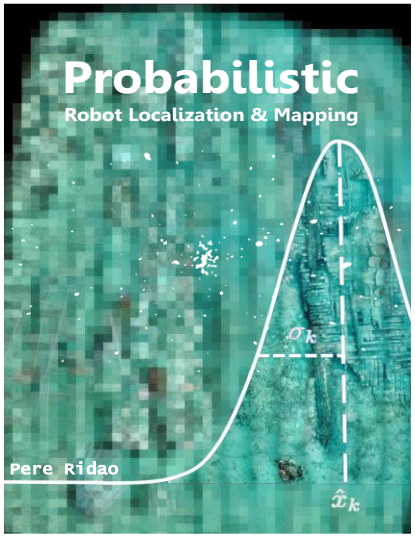
http://www.youtube.com/watch?v=asyhZUuzENw&feature=relmfu

UdG

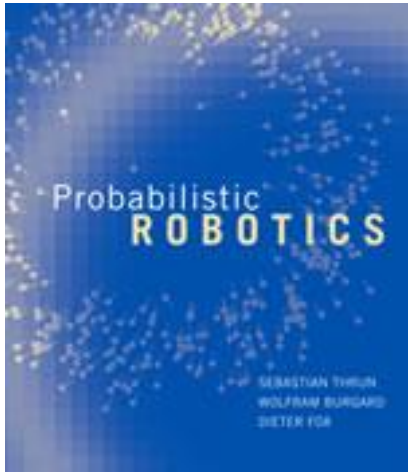
Outline

1. Introduction
2. Kalman Filter
3. Extended Kalman Filter
4. Map Based Localization
5. Feature Based EKF SLAM
6. Pose Based EKF SLAM

Bibliography



Probabilistic Robot Localization & Mapping
Lecture Notes
Pere Ridao.



Probabilistic Robotics
Sebastian Thrun, Wolfram Burgard and Dieter Fox
The MIT-Press
ISBN-10: 0-262-20162-3

UdG

Labs

1. Lab: Differential Robot Simulation
2. Lab: Dead Reckoning
3. Lab: Kalman Filter
4. Lab: EKF
5. Lab: EKF Map Based Localization
6. Feature Based EKF SLAM

Labs

- **Team work (2 students)**

- **Procedure:**
  - **Pre-Lab session:**
    - Read the exercise description
    - Prepare for the lab and potentially solve preliminary questions.
    - The pre-lab may include a submission too.
  - **Lab session:**
    - Perform your work in teams.
  - **Post-Lab session:**
    - Write a report.

- **Evaluation:** 25% Report + 75 % Lab Quiz (either written or oral)

- **Assistence is mandatory.** **Not assisting** with due justification will be **evaluated with 0**.
- **Submission should be on time:** Every day **delay will reduce 1 point** the potential mark.
- **Access to the lab beyond assigned hours will be granted** when there is no teaching in the lab.

About the Use of AI

AI may be used to:

- Clarify theoretical concepts related to the laboratory.
- Explain scientific or engineering principles.
- Improve the grammar, spelling, and readability of laboratory reports.
- Help understand feedback provided by the instructors.
- You can use it for generate small code snippets but you are completely responsible for the result. You must be able to explain any function or variable in your code. You must also be able to take modifications of the code according to requests by the evaluator.
- You will be examined about the contens of the code and about how it performs.

- **If AI is used for any permitted purpose its use should be acknowledged** briefly in the submitted report.

During Evaluation Activities

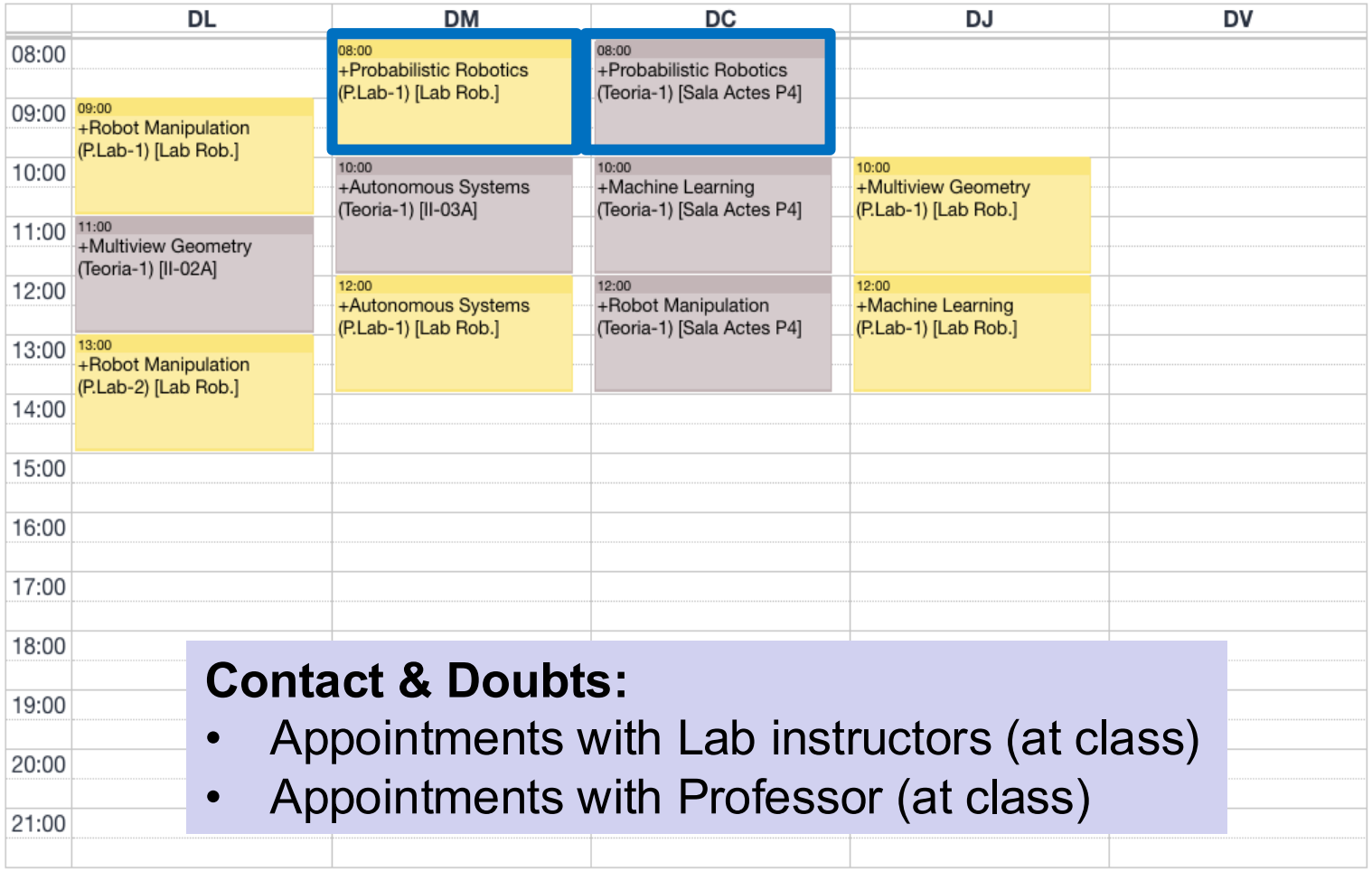
Students must enter the classroom where the assessment activity is to take place with all communication devices — mobile phones, computers, tablets, smartwatches, etc. — switched off and kept inside their backpacks/bags. Failure to comply with this rule will result in a grade of 0 for the activity, as well as the implementation of the actions described in Article 21 of the UdG regulations governing student assessment and grading processes.

If, during the correction process of the assessment activity, the lecturer determines the existence of possible fraud, they reserve the right to validate the grade obtained using the assessment methodology they deem appropriate.

Activities & Evaluation

• *Labs*		*30 %. > 5*
• *Partial Examination*	*35 % > 4 makes average*	
• *Final Examination*	*35 % > 4 makes average*	*70 % > 5*

Schedulling



Contact & Doubts:

- Appointments with Lab instructors (at class)
- Appointments with Professor (at class)